KrishiGuide

AI Crop Care, Powered by NASA Earth Data

Software Requirements Specification

KrishiGuide — NASA Farm Navigators Edition

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1. Introduction

1.1 Problem Statement

Smallholder farmers around the world, and especially across South and Southeast Asia, continue to lose significant yield every year to crop disease, unpredictable weather, and inefficient irrigation decisions. Farmers face several recurring challenges, regardless of country:

- Late detection of crop disease due to lack of access to agronomists
- Irrigation and fertilizer decisions based on guesswork rather than real environmental data
- No easy access to satellite-based soil moisture, rainfall, or vegetation-health information
- Vulnerability to drought and flood events that could be anticipated with Earth observation data
- Limited digital tools that combine disease diagnosis with environmental context

These gaps lead to reduced crop yield, wasted water and inputs, and delayed responses to plant disease outbreaks — a global problem that NASA's freely available, worldwide Earth-observation data is well suited to help solve.

1.2 Proposed Solution

We propose KrishiGuide, an AI-powered mobile/web application that combines on-device crop disease detection with NASA Earth observation data to give any farmer, anywhere, a single, actionable "Farm Intelligence" view. Because NASA POWER, SMAP, GPM, and MODIS/VIIRS datasets provide global coverage, the same architecture works for a rice farmer in Bangladesh, a maize farmer in Kenya, or a wheat farmer in India. The system will enable:

- AI-based detection of crop disease from a leaf photograph
- Retrieval of NASA satellite data (soil moisture, rainfall, temperature, vegetation index) for the farmer's exact GPS location, anywhere on Earth
- Combined decision support that merges disease diagnosis with environmental risk (e.g. irrigation alerts, disease-favorable-weather warnings)
- Localized, easy-to-understand action plans, translatable into the farmer's own language
- Historical trend tracking of soil and weather conditions per plot

1.3 Scope and Global Applicability

KrishiGuide is being built for the NASA Space Apps Challenge theme "NASA Farm Navigators," which calls for solutions that use NASA open data to support agricultural decision-making anywhere in the world. The core system is location-agnostic — it works wherever NASA's global datasets have coverage, which is effectively everywhere. The project's scope covers:

- Global architecture: works for any crop and any GPS location, not tied to a single country's boundaries
- Bangladesh as the initial pilot and demo region — chosen for accessible field validation, a large smallholder farming population, and the team's local agronomy contacts
- Integration of NASA POWER, SMAP soil-moisture, GPM rainfall, and MODIS/VIIRS NDVI datasets, all of which provide worldwide coverage
- Early warning for drought, flood, and disease-favorable weather conditions, applicable to any climate zone
- A demonstrable prototype showing how NASA data changes a real farming decision, as required by the challenge's judging criteria, with a clear path to scale to other countries and regions

2. Overall Description

2.1 Product Perspective

KrishiGuide is a stand-alone, farmer-facing mobile-first application. It acts as a bridge between the farmer, an on-device/cloud AI disease-detection model, and NASA's open Earth-observation data — with no separate officer or admin portal required for the farmer to get value from the app. The system interfaces with:

- NASA POWER API and NASA Earthdata (soil moisture, rainfall, temperature, NDVI)
- A trained image-classification model for crop disease detection
- SMS/push notification services for alerts in low-connectivity areas

2.2 Product Features

- Farmer Registration & Profile Management
- AI Crop Disease Detection from photograph
- NASA Farm Intelligence dashboard (soil moisture, rainfall, temperature, NDVI)
- Combined Action Plan generation (disease + environment)
- Irrigation and drought/flood early-warning alerts
- Historical trend graphs per plot/location
- Multilingual support (Bangla and English)
- Offline-friendly, low-bandwidth data views for rural connectivity

2.3 User Class & Characteristics

KrishiGuide has a single user class — the Farmer — who is the sole person interacting with the application. There is no separate officer or administrator front-end; any backend maintenance is handled by the development team directly on the server and does not form part of the app's user-facing design.

User Role	Description / Skill Level
Farmer	Uploads crop photos, views NASA farm intelligence and action plans; basic mobile skills, may need Bangla-language and voice support.

2.4 Operating Environment

- Hardware: Android smartphone with a camera (primary).
- Operating Systems: Android 8+.
- Network: Functions on 2G/3G with low-bandwidth mode; full features on 4G/Wi-Fi.
- Database: Firebase / MySQL for structured data; cloud storage for uploaded images.
- Third-Party Services: NASA POWER API, NASA Earthdata, SMS gateway.

2.5 Design & Implementation Constraints — Technology Stack

The table below lists the specific languages, frameworks, and APIs planned for KrishiGuide, chosen to match the team's existing skill set (Python, JavaScript/React, Firebase, MySQL/SQLite) and the free, well-documented nature of NASA's public APIs.

Layer	Technology / API
Mobile App (Frontend)	React Native (JavaScript) — single codebase for Android and iOS
Backend / API Server	Python 3 with Flask (REST API)
AI / Disease Detection Model	Python, TensorFlow / Keras (CNN image classifier); TensorFlow Lite for on-device/mobile inference
Database	Firebase Firestore (real-time farmer/plot data) with MySQL or SQLite for structured relational/reporting data
Authentication	Firebase Authentication (phone-number OTP login)
Image Storage	Firebase Cloud Storage (uploaded leaf/crop photos)
Push Notifications	Firebase Cloud Messaging (FCM)
SMS Alerts	Twilio API or a local telecom SMS gateway (for low-connectivity/feature-phone-adjacent alerts)
Location / Maps	Device GPS + Google Maps Geocoding API (for plot location and reverse-geocoding)
NASA Soil Moisture	NASA SMAP (Soil Moisture Active Passive) via NASA Earthdata / NASA-USDA Global Soil Moisture data
NASA Rainfall	NASA GPM IMERG (Global Precipitation Measurement) via NASA Earthdata
NASA Temperature & Weather	NASA POWER API (Prediction of Worldwide Energy Resources) — free, no-auth REST API
NASA Vegetation Health (NDVI)	NASA MODIS / VIIRS (via NASA Earthdata / AppEEARS or Land Processes DAAC)
Version Control	Git & GitHub
Hosting / Deployment	Cloud platform such as Render, Railway, or AWS/GCP free-tier for the Flask backend; Firebase Hosting for any companion web pages
Dev Tools	VS Code, Postman (API testing)

- NASA data must be cached locally (in Firestore/SQLite) to reduce repeated API calls and support low-connectivity use.
- Mobile app must remain usable on entry-level devices with 2GB RAM.

2.6 User Documentation

- In-app guided tutorial for first-time farmers (icon-based, minimal text)
- FAQ and voice-guided help for low-literacy users

2.7 Assumptions & Dependencies

- Farmers have access to a basic smartphone with a working camera.
- NASA POWER / Earthdata APIs remain publicly available and free to access.
- Location (GPS or manually entered) is reasonably accurate for satellite-data lookups.
- Local weather patterns are broadly reflected by the chosen NASA datasets.

3. Specific Requirements

3.1 Functional Requirements

3.1.1. User Registration and Login

- Farmers can register using phone number or email.
- Secure login with password and optional OTP-based two-factor authentication.

3.1.2. AI Crop Disease Detection

- Farmer can upload or capture a leaf/crop photo.
- System returns disease name, confidence score, and recommended treatment.
- If confidence is low, the app tells the farmer directly and suggests retaking the photo (closer, better light, different leaf) rather than routing to a human reviewer.

3.1.3. NASA Farm Intelligence Retrieval

- System fetches soil moisture, rainfall, temperature, and NDVI for the farmer's location from NASA POWER/Earthdata.
- Data is cached locally and refreshed on a daily schedule.
- Farmer can view a simplified summary (Low/Moderate/High) as well as raw trend graphs.

3.1.4. Combined Action Plan Generation

- System merges disease-detection output with NASA environmental data.
- Rule-based (and later ML-based) engine generates an actionable recommendation, e.g. irrigation advice or disease-favorable weather warning.
- Action plan is shown in plain, non-technical Bangla/English language.

3.1.5. Alerts & Notifications

- System sends push/SMS alerts for drought risk, heavy-rain forecasts, or disease-favorable conditions.
- Farmer can configure alert preferences (crop type, frequency).

3.1.6. Plot & History Management

- Farmer can register multiple plots with crop type and location.
- System stores history of disease detections and environmental readings per plot.
- Farmer can view trend charts over the growing season.

3.2 Non-Functional Requirements

3.2.1 Performance

- Disease detection result must return within 5 seconds on a stable connection.
- NASA data dashboard must load within 3 seconds using cached data.

3.2.2 Security

- All farmer data and images must be encrypted in transit and at rest.
- Farmer accounts are protected with secure authentication and OTP verification.

3.2.3 Usability

- Interface must use icons and minimal text for low-literacy users.
- Full Bangla-language support, including voice prompts where feasible.

3.2.4 Scalability

- Backend must handle growth from pilot villages to district-wide (and eventually multi-country) rollout without redesign.

3.2.5 Availability

- Target 99% uptime; core disease-detection and cached NASA data must work with intermittent connectivity.

3.2.6 Maintainability

- Modular codebase separating the AI model, NASA data layer, and application logic for independent updates.

3.2.7 Compatibility

- Compatible with Android 8+ devices.

3.2.8 Data Integrity

- NASA data pulls and AI results are timestamped and versioned to avoid conflicting readings.

4. Feasibility Study

4.1 Technical Feasibility

- Cloud-hosted backend with a farmer-facing mobile app; no separate officer/admin front-end is required for MVP.
- NASA POWER API is free, public, and does not require complex authentication — well suited for a challenge-timeline prototype.
- AI disease-detection model can be trained on existing open crop-disease image datasets (e.g. PlantVillage) and fine-tuned for local crops.

4.1.1 Software Feasibility

- Required software: Python/Flask backend, React Native frontend, TensorFlow/Keras for the AI model, and client libraries for NASA POWER, SMAP, GPM, and MODIS/VIIRS APIs (all free and require no complex authentication for research/prototype use).
- Challenge: aligning NASA satellite data resolution (often multi-km grid) with individual farm plots.

4.1.2 Hardware Feasibility

- Entry-level Android smartphones for farmers; standard cloud server infrastructure for backend and model hosting.
- Challenge: ensuring smooth camera-based capture in varied outdoor lighting conditions.

4.2 Human Resource Feasibility

- Required personnel: software engineers, an ML engineer, an agronomist consultant for validation, and local field partners.
- Challenge: building trust with farming communities and providing simple onboarding.

4.3 Economic Feasibility

4.3.1 Identification of Costs and Benefits

- Initial costs: app/platform development, cloud infrastructure, AI model training, community outreach and training.
- Ongoing costs: cloud hosting, SMS/notification services, maintenance, dataset updates.
- Estimated benefits: reduced crop loss from earlier disease detection, more efficient irrigation and water use, reduced dependence on guesswork.

4.3.2 Financial Viability

A detailed cost-benefit and ROI analysis will be prepared once pilot-village deployment costs are confirmed; at prototype stage the primary investment is development time and NASA data integration effort, both of which are low-cost given NASA's open-data policy.

4.4 Operational Feasibility

- Can be piloted through agricultural extension offices, NGOs, and university outreach programs.
- Fits naturally into the NASA Farm Navigators challenge's goal of using NASA data for real agricultural decisions.

4.5 Legal and Social Feasibility

- Complies with local data-protection expectations for farmer personal data in each deployment country, starting with Bangladesh.
- NASA data usage complies with NASA's open-data terms of use, which permit global redistribution and application development.
- Social acceptability supported through community demonstrations and local-language design.

5. Requirements Elicitation

5.1 Technique Used

- Interviews
- Questionnaires
- Brainstorming
- Document Analysis

5.2 Application of Techniques

Interviews

Informal interviews were conducted with local farmers and an agricultural extension contact to understand pain points with current disease-identification and irrigation-decision practices, including delayed diagnosis and reliance on guesswork for watering schedules.

Questionnaires

A short survey concept was drafted to gather farmer preferences on alert frequency, preferred language, and the type of environmental information (soil moisture vs. rainfall vs. temperature) considered most useful.

Brainstorming

Team brainstorming sessions mapped the combined "Disease AI + NASA Environmental Data" workflow and identified the Action Plan screen as the single most valuable feature for demonstrating the NASA Farm Navigators theme.

Document Analysis

The team reviewed NASA Space Apps Challenge documentation for "NASA Farm Navigators," NASA POWER/Earthdata API documentation, and existing KrishiGuide feature specifications to ensure alignment between the existing app and the new NASA-data features.

6. User Profile

6.1 User Profile — Farmer

Characteristic	Notes on characteristic	Requirement implied
Type of user	Smallholder farmer, primary end user	Simple, icon-driven, low-literacy-friendly interface
Age range	20–65	Large fonts, minimal text, voice support
Frequency of use	Daily during growing season	Fast, low-bandwidth-friendly interface
Mandatory/discretionary	Discretionary but high-value during disease outbreaks	Must work reliably when needed most
Computer experience	Basic smartphone use only	Guided onboarding, no jargon
Uses	Upload crop photo, view NASA farm intelligence, receive action plan	Combined, single-screen recommendations
Education	Varies, often minimal formal education	Bangla language, icons over text
Goals	Protect crop yield, avoid wasted water/inputs	Clear, actionable recommendations
Language skills	Bangla (regional dialects), limited English	Full Bangla support
Other systems used	May use basic weather apps or none	Optional integration with local weather services

7. Use Case Modeling

7.1 Use Case Overview

The Farmer is the only primary actor in the system, with the AI Model and NASA Data Service acting as supporting (secondary) actors. Key use cases include Register, Login, Detect Disease, Fetch NASA Farm Data, Generate Action Plan, Manage Plot History, and Send Alert.

7.2 Use Case Description

7.2.1. Use Case: Register User

Goal: New users create an account by providing required information.

Preconditions: Registration page is accessible.

Primary Actors: Farmer

Trigger: User taps "Register" and fills the form.

Main Success Scenario:

1. User accesses registration page.
2. User fills in name, phone/email, password, and location.
3. System validates input.
4. System stores user details.
5. Confirmation message is shown.

Alternative Flows: If phone/email already exists, system shows an error.

Quality Requirements: Password must be encrypted; registration completes within 3 seconds.

7.2.2. Use Case: Login User

Goal: Farmer logs into the system using valid credentials.

Preconditions: Farmer is registered.

Primary Actors: Farmer

Trigger: Farmer taps "Login".

Main Success Scenario:

1. Farmer enters phone/email and password.
2. System validates credentials.
3. Session is started.
4. Farmer is redirected to the home dashboard.

Alternative Flows: Invalid credentials → system displays error and limits attempts.

Quality Requirements: Login completes within 2 seconds; lockout after 5 failed attempts.

7.2.3. Use Case: Detect Crop Disease

Goal: Farmer uploads a crop photo and receives an AI-based disease diagnosis.

Preconditions: Farmer is logged in; camera or gallery access is available.

Primary Actors: Farmer

Secondary Actors: AI Model

Trigger: Farmer selects "Detect Disease" and captures/uploads a photo.

Main Success Scenario:

1. Farmer captures or uploads a leaf/crop photo.
2. System sends the image to the AI model.
3. AI model returns disease name and confidence score.
4. System stores the result under the farmer's plot history.
5. Result and recommended treatment are displayed.

Alternative Flows: Low confidence (<70%) → app tells the farmer directly and suggests retaking the photo.

Quality Requirements: Result returned within 5 seconds; image securely stored.

7.2.4. Use Case: Fetch NASA Farm Intelligence

Goal: Retrieve soil moisture, rainfall, temperature, and NDVI data for the farmer's plot location.

Preconditions: Farmer's plot location is set (GPS or manual entry).

Primary Actors: Farmer

Secondary Actors: NASA Data Service (POWER/Earthdata)

Trigger: Farmer opens the "NASA Farm Intelligence" screen.

Main Success Scenario:

1. System checks for a valid cached reading for the location.
2. If stale, system requests fresh data from NASA POWER/Earthdata API.
3. System converts raw data into simplified Low/Moderate/High indicators.
4. Data is displayed with trend graphs.

Alternative Flows: NASA API unavailable → system serves last cached data with a "last updated" notice.

Quality Requirements: Dashboard loads within 3 seconds using cache; data refreshed at least daily.

7.2.5. Use Case: Generate Combined Action Plan

Goal: Produce a single actionable recommendation combining disease diagnosis and NASA environmental data.

Preconditions: A recent disease-detection result and NASA data reading exist for the plot.

Primary Actors: Farmer

Secondary Actors: AI Model, NASA Data Service

Trigger: System automatically generates the plan after a disease detection or scheduled data refresh.

Main Success Scenario:

1. System retrieves the latest disease result and NASA data for the plot.
2. Decision-support logic evaluates combined risk (e.g. low soil moisture + disease-favorable temperature).
3. System generates a plain-language recommendation (e.g. irrigate, spray before rain, monitor).
4. Action plan is displayed and optionally sent as an alert.

Alternative Flows: Insufficient data → system shows a partial plan with a note on missing information.

Quality Requirements: Plan generated within 2 seconds of underlying data being available.

7.2.6. Use Case: Send Alert / Notification

Goal: Notify farmers of drought risk, heavy rainfall, or disease-favorable conditions.

Preconditions: Alert thresholds are configured; NASA data indicates a risk condition.

Primary Actors: System

Secondary Actors: Farmer

Trigger: NASA data refresh crosses a configured risk threshold.

Main Success Scenario:

1. System evaluates the latest NASA data against thresholds.
2. System composes an alert message in the farmer's preferred language.
3. Alert is sent via push notification or SMS.
4. Delivery is confirmed and logged.

Alternative Flows: Delivery fails → system retries and logs the failure.

Quality Requirements: Alert sent within 10 seconds of threshold breach being detected.

8. Requirement Prioritization

8.1 Technique Used

MoSCoW (Must have, Should have, Could have, Won't have this time)

8.2 Prioritized List

Req. ID	Description	Priority
FR001	User registration	Must Have
FR002	User login	Must Have
FR003	AI crop disease detection	Must Have
FR004	Fetch NASA farm intelligence (soil moisture, rainfall, temp, NDVI)	Must Have
FR005	Generate combined action plan	Must Have
FR006	Plot & history management	Must Have
FR007	Alerts & notifications	Should Have
FR008	Multilingual (Bangla) support	Should Have
FR009	Offline / low-bandwidth mode	Could Have
FR010	Voice-guided assistance	Could Have

9. Software Requirements Specification

System Name: KrishiGuide — NASA Farm Navigators Edition

FR001 User Registration and Login

Description: Farmers register and log in securely using a phone number or email. OTP-based verification is supported for farmers with only a phone number.

Stakeholders: Farmer

FR002 AI Crop Disease Detection

Description: Farmers can capture or upload a photo of a crop leaf; the system returns a disease name, confidence score, and recommended treatment.

Stakeholders: Farmer, AI Model

FR003 NASA Farm Intelligence Retrieval

Description: The system retrieves soil moisture, rainfall, temperature, and NDVI data for the farmer's registered plot location from NASA POWER/Earthdata and presents it as simplified indicators and trend graphs.

Stakeholders: Farmer, NASA Data Service

FR004 Combined Action Plan

Description: The system merges the disease-detection result with current NASA environmental data to generate a single, plain-language recommendation.

Stakeholders: Farmer, AI Model, NASA Data Service

FR005 Plot & History Management

Description: Farmers can register multiple plots and view historical disease and environmental data trends per plot.

Stakeholders: Farmer

FR006 Alerts & Notifications

Description: The system sends push/SMS alerts when NASA data indicates drought, heavy rainfall, or disease-favorable weather.

Stakeholders: Farmer, System

FR007 Multilingual Support

Description: The interface and all recommendations are available in Bangla and English.

Stakeholders: Farmer

10. Traceability Matrix

SN	Req. ID	Requirement	Scenario	Test ID	Test Description	Result
01	FR001	Register User	New user registers account	TC001	Enter registration information	Pass
02	FR001	Register User	New user registers account	TC002	Submit registration form	Pass
03	FR002	Detect Disease	Farmer uploads leaf photo	TC003	Upload valid leaf image	Pass
04	FR002	Detect Disease	Low-confidence result flagged	TC004	Upload unclear/blurred image	Pass
05	FR003	Fetch NASA Data	System fetches soil/rainfall data	TC005	Request data for valid location	Pass
06	FR003	Fetch NASA Data	API temporarily unavailable	TC006	Serve cached data on API failure	Pass
07	FR004	Generate Action Plan	Combine disease + environment data	TC007	Generate plan for low soil moisture case	Pass
08	FR005	Manage Plot History	Farmer views trend history	TC008	View 30-day soil moisture trend	Pass
09	FR006	Send Alert	Drought risk threshold breached	TC009	Trigger and deliver alert	Pass
10	FR007	Multilingual Support	Farmer switches language	TC010	Toggle Bangla/English interface	Pass
11	FR002	Detect Disease	Low-confidence retake flow	TC011	Prompt farmer to retake unclear photo	Pass

11. Prototype Design

11.1 Tool Used

Figma (wireframes to be finalized during the NASA Space Apps Challenge build phase).

11.2 Planned Screens

- Splash / Register / Login — simple entry with phone-number OTP login.
- Home — at-a-glance weather and soil summary card, quick actions (Detect Disease, Farm Intelligence, History).
- Disease Detection — camera/upload button, AI result with confidence score and treatment advice.
- NASA Farm Intelligence — soil moisture, rainfall trend, temperature, and NDVI shown as simple Low/Moderate/High indicators plus graphs.
- Action Plan — a single combined card merging disease result and environmental risk into one recommendation.
- History — past disease detections and environmental trends per plot.

Note: detailed Figma screenshots will be attached once the visual design pass for the challenge submission is complete.

12. Conclusion

KrishiGuide — NASA Farm Navigators Edition combines an existing AI crop-disease detection engine with NASA's open Earth-observation data to give smallholder farmers a single, actionable view of both plant health and environmental risk. By integrating soil moisture, rainfall, temperature, and vegetation-health data with disease diagnosis, the system moves beyond a simple disease detector into a genuine agricultural decision-support tool.

The system directly addresses the NASA Farm Navigators challenge goal of using NASA data to improve real farming decisions anywhere in the world, while remaining practical for low-connectivity rural regions through low-bandwidth support, multilingual design, and a simple combined Action Plan screen. Bangladesh serves as the initial pilot and demo region, but the same architecture is designed to scale to any country covered by NASA's global datasets. Future work includes expanding the environmental decision-support logic from rule-based to machine-learning-based, and piloting the system with farming communities and agricultural extension partners in additional regions.